Assignment 0: Introduction
Advanced Topics in Machine Learning

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https://github.com/sramiza/ATML-Assignment-0.git

Abstract

This report details the implementation and analysis of foundational deep learning concepts, including convolutional transfer learning, self-attention mechanisms in Vision Transformers, multimodal contrastive learning, and variational autoencoders.

1 Task 1: Inner Workings of ResNet-152

1.1 Baseline Setup

Training a 60-million parameter model like ResNet-152 entirely from scratch on a relatively small dataset such as CIFAR-10 is both unnecessary and impractical: the model carries a very high risk of overfitting on a dataset of CIFAR-10's size, and training such a deep backbone from random initialization demands substantially more compute and wall-clock time than fine-tuning already-learned representations. The foundational visual features, such as edges and textures, have already been robustly learned on ImageNet. By freezing the network backbone and only training the final classification head, we demonstrate the universal transferability of these hierarchical features. During our baseline training, the model successfully adapted to the new 10-class distribution, with the training loss steadily decreasing to 0.5243 and validation accuracy reaching 44.56% over three epochs.

1.2 Residual Connections in Practice

To observe the practical impact of residual connections, we conducted an ablation study by severing the skip connections (`out += identity`) in multiple selected residual blocks (specifically within `layer3` and `layer4`). Without these shortcuts, the network suffers from a localized vanishing gradient problem, as the gradients are forced through the heavy convolutional layers without an uninterrupted path for backpropagation. Consequently, the modified model's training stalled, plateauing at a loss of 2.02 and a severely degraded validation accuracy of 31.79%.

1.3 Feature Hierarchies and Representations

To visualize how the network transforms raw pixels into semantic representations, we extracted feature embeddings from three different depths of the baseline network and projected them into 2D space using t-SNE. As shown in Figure 1, the Early Layer (Layer 1) exhibits zero class separability, as it only captures low-level visual primitives shared across all images. The Middle Layer (Layer 3) begins to form loose local clusters as the network detects mid-level structures. Finally, the Late Layer (AvgPool) resolves into sharp, distinct clusters, demonstrating that the deep convolutional backbone successfully separates the data into high-level semantic classes.

Preprint.

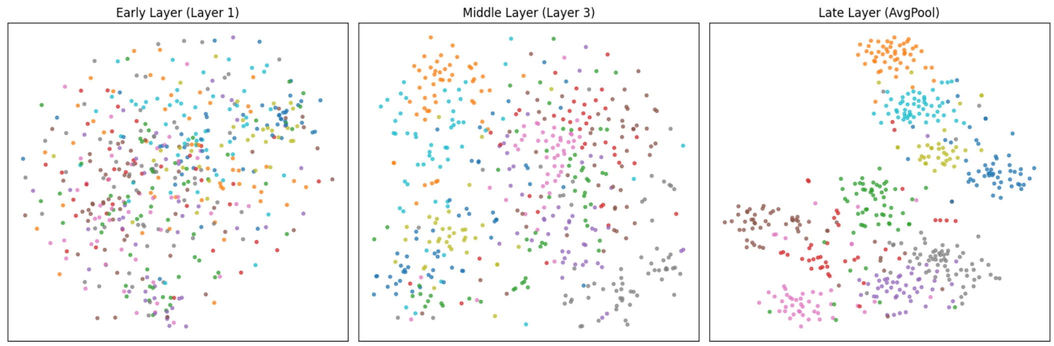

Figure 1: t-SNE visualization of CIFAR-10 test set embeddings extracted from Layer 1, Layer 3, and AvgPool of ResNet-152.

1.4 Transfer Learning and Generalization

We conducted a rapid 1-epoch comparative analysis to evaluate the trade-offs between computational cost and model accuracy across different transfer learning strategies. The results are summarized in Table 1. Training the full backbone from scratch yielded the worst performance (48.13%) while consuming the most compute time (1349 seconds). Leveraging pretrained weights and training only the head provided a massive efficiency boost, cutting compute time nearly by a third while achieving 82.41% accuracy. However, fine-tuning the final convolutional block alongside the head proved to be the optimal strategy; it incurred a minimal computational penalty over the head-only approach while delivering the highest accuracy at 90.98%, allowing the model to adapt its highest-level semantic representations specifically to the CIFAR-10 dataset.

Table 1: Comparison of Transfer Learning Strategies on CIFAR-10 (1 Epoch)

Training Strategy	Validation Accuracy	Compute Time (s)
Random Initialization (Full Backbone)	48.13%	1349.61
Pretrained Weights (Head Only)	82.41%	490.51
Pretrained Weights (Layer 4 + Head)	90.98%	543.03

2 Task 2: Understanding ViT

2.1 Pre-trained Inference

The pre-trained ViT-B/16 successfully identified the test images, classifying the first image as a 'Samoyed' with 79.97% confidence and the second as a 'sports car' with 55.78% confidence. This establishes a strong baseline inference capability using the pre-trained weights without any task-specific fine-tuning.

2.2 Self-Attention Visualizations and Limitations

To understand the spatial focus of the Vision Transformer, we extracted the self-attention weights from the final encoder layer corresponding to the `[CLS]` token. As observed in our generated heatmaps (Figure 2), extracting the raw attention maps from only the final layer does not yield a perfect segmentation of the primary object. For instance, the attention weights for the Samoyed heavily activate on the background vegetation, while the sports car's attention focuses on the windshield and surrounding environment. This behavior is expected in raw ViT architectures. In deep Transformers, earlier layers aggregate the primary spatial features. By the final layer, the `[CLS]` token often attends to peripheral contextual clues to finalize the scene classification. Furthermore, simply averaging the multi-head attention weights at the final stage introduces noise. Achieving precise object segmentation typically requires techniques like Attention Rollout, which recursively computes attention across all layers rather than isolating the final block.

Figure 2: Self-attention heatmaps extracted from the final layer of ViT-B/16 for the `[CLS]` token.

Conceptually, compared to typical CNN interpretation methods like CAM or Grad-CAM, which rely on post-hoc gradient approximations backpropagated onto convolutional feature maps, Transformers offer a distinct advantage by providing built-in attention distributions that map directly to the input tokens without requiring any additional gradient computation or architectural modification. Furthermore, analyzing individual attention heads reveals specialization: some heads focus heavily on the foreground subject and closely trace its outline, while others exhibit unusual behavior by specializing entirely on background or irrelevant contextual cues. We determine this specialization by visualizing the per-head attention maps separately (rather than averaging across heads) and observing that different heads consistently activate over different spatial regions across images, indicating that each head has learned to attend to a distinct type of visual pattern.

2.3 Robustness to Patch Masking and Occlusion

To evaluate the structural robustness of the Vision Transformer, we conducted an occlusion experiment by artificially masking out 50% of the image patches. Remarkably, despite losing half of the visual data, the ViT maintained its predictive capabilities (Figure 3). The heavily masked dog was classified as a 'Great Pyrenees' (a visually similar breed to the baseline 'Samoyed') with 49.4% confidence, and the masked vehicle was still accurately identified as a 'sports car' with 47.2% confidence. This

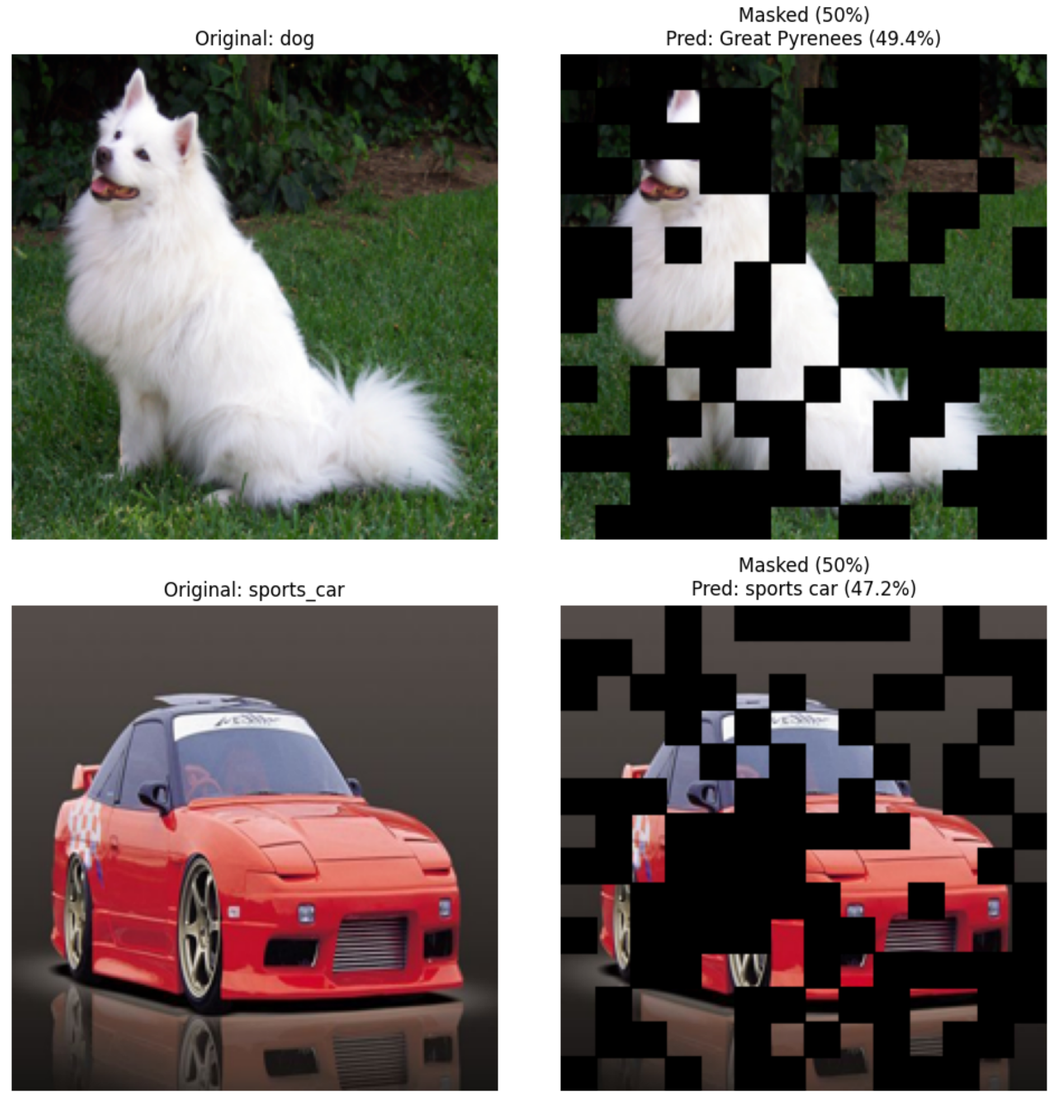

Figure 3: ViT-B/16 inference results after masking 50% of the input patches with black occlusions.

resilience highlights a fundamental advantage of the self-attention mechanism over standard convolutions. While CNNs rely on rigid, localized receptive fields that degrade heavily when contiguous pixel structures are destroyed, the ViT processes the image as a global sequence of independent patch tokens. The multi-head self-attention mechanism dynamically computes relationships across the entire image space. When severe occlusion occurs, the network can flexibly adjust its attention weights to route contextual information between the surviving, non-adjacent patches, enabling it to accurately reconstruct the semantic meaning of the scene from highly fragmented data.

Comparing random masking against structured masking reveals an important distinction. Under random masking, the ViT remains robust because the surviving patches are scattered across the entire image, so some sample of every semantically important region (head, legs, torso, etc.) is likely to remain visible somewhere in the sequence, giving self-attention enough scattered context to reconstruct the object's identity. Under structured masking, such as zeroing out a contiguous block in the center of the image, the network suffers a more severe drop in accuracy. This is because structured occlusion can eliminate an entire localized structural feature at once (e.g., the whole face or body of the subject) rather than merely thinning it out, leaving no surviving patches within that region for self-attention to draw on. This confirms that the ViT's robustness comes specifically from the spatial *distribution* of surviving information, not merely from the total fraction of visible pixels.

2.4 CLS Token vs. Mean-Pooled Patch Tokens

When comparing pooling methods for downstream classification via linear probes, empirical results demonstrated that training on the mean of all patch tokens outperformed training strictly on the `[CLS]` token. Specifically, the `[CLS]` token probe achieved a test accuracy of 37.00%, whereas the mean-pooled probe reached 40.80%. This occurs because the semantic information in a Vision Transformer is distributed across the entire sequence of patch embeddings. While the `[CLS]` token serves as a global summary rigidly tied to a specific pre-training classification objective, mean pooling aggregates a richer, more comprehensive representation of the full image context.

3 Task 3: Contrastive Learning and CLIP Model

3.1 Zero-Shot Classification

Unlike traditional convolutional networks such as ResNet, which are constrained to a fixed number of predetermined classes (e.g., the 1,000 ImageNet categories), the CLIP (Contrastive Language-Image Pretraining) model aligns visual and textual representations within a shared latent vector space. This architectural shift enables zero-shot classification, allowing the model to accurately categorize images based on open-ended natural language prompts without requiring task-specific fine-tuning or labeled training data. To demonstrate this capability, we evaluated a pre-trained `clip-vit-base-patch32` model using an unconstrained image of a SpaceX rocket launch. We provided the model with four custom, diverse text prompts. The results of the zero-shot inference are as follows:

- **'a photo of a space rocket':** 99.61%
- **'an astronaut riding a horse on mars':** 0.39%
- **'a person walking a dog in the park':** 0.00%
- **'a standard portrait of a man':** 0.00%

Despite the lack of explicit training on a "space rocket" class, the model successfully matched the visual data to the correct semantic text embedding with 99.61% confidence. This confirms the efficacy of contrastive pre-training in generalizing across diverse and entirely novel visual concepts.

3.2 Multimodal Similarity Heatmaps

To visualize how CLIP bridges the modality gap between vision and language, we generated patch-level similarity heatmaps. We divided an input image into a grid of patches, independently computed the visual embedding for each patch, and calculated the cosine similarity against the text embeddings for two distinct prompts: 'a dog' and 'a cat'. As shown in Figure 4, the model exhibits highly

Figure 4: Patch-level cosine similarity heatmaps demonstrating CLIP's spatial localization of text concepts.

accurate spatial localization despite never being explicitly trained for bounding-box object detection. When queried with the prompt 'a cat', the highest similarity scores (indicated by the high-activation regions) precisely localize over the feline's face and distinct features. Conversely, when queried with 'a dog'—an object absent from the scene—the model fails to find a meaningful semantic match on the primary subject, resulting in scattered activations relegated primarily to the background context. This confirms that CLIP's contrastive pre-training successfully enforces a structurally aware, shared latent space where visual features and semantic text are tightly aligned.

3.3 CLIP on STL-10 and the Modality Gap

3.3.1 Zero-Shot Classification

The evaluation of CLIP on the STL-10 dataset across four distinct prompting strategies yielded the following top-1 accuracies:

- **Plain labels:** 0.9625
- **Photo prompts:** 0.9736
- **Detailed prompts:** 0.9674
- **Contextual prompts:** 0.9645

The results indicate that the choice of prompt influences zero-shot classification accuracy in CLIP. The superior performance of "Photo prompts" supports the hypothesis that CLIP's vision-language training distribution is biased toward naturalistic captions from internet data, making the model more sensitive to prompts that resemble such captions. Detailed prompts, despite containing more descriptive information, did not yield a significant accuracy boost, suggesting that CLIP prioritizes semantic alignment over linguistic richness.

3.3.2 Exploring and Bridging the Modality Gap

To investigate the geometric relationship between visual and textual representations, we extracted image and label embeddings for a subset of STL-10 samples. As observed in the t-SNE visualization (Figure 5, left panel), the shared embedding space exhibits a pronounced modality gap, where embeddings from different modalities are visibly separated in the latent space. To align the modalities, an orthogonal Procrustes transform was utilized. Given the image embeddings representing one set and the text embeddings representing the other, the objective is to find an orthogonal matrix that minimizes the Frobenius norm between them. Post-alignment visualization (Figure 5, right panel)

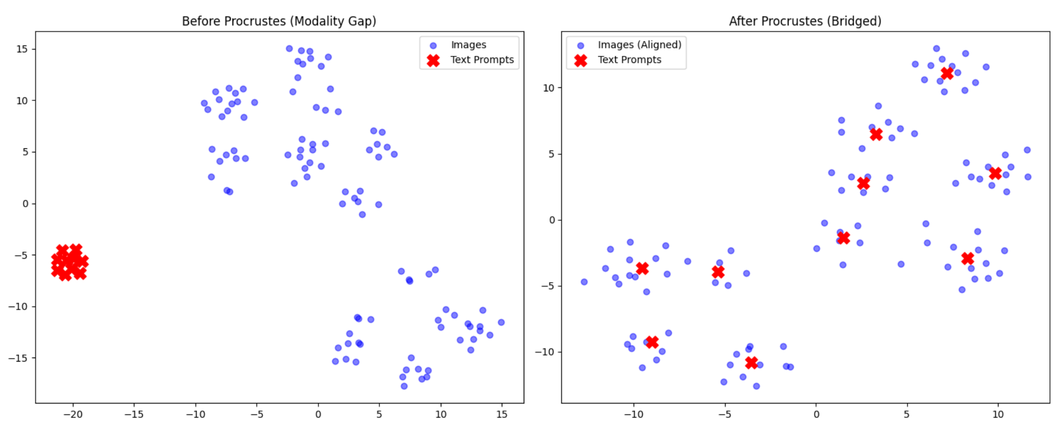

Figure 5: t-SNE visualization of the modality gap before (left) and after (right) Orthogonal Procrustes alignment.

demonstrates that the text prompts were successfully rotated directly into the geometric centers of their corresponding image clusters, effectively bridging the physical modality gap. However, the recomputed zero-shot classification accuracy using the aligned embeddings yielded a score of 0.9685. Because zero-shot classification relies on cosine similarity—which is invariant to global orthogonal rotations—the Procrustes transformation mathematically minimizes the Euclidean distance between the two domains without fundamentally altering the angular separability of the classes.

4 Task 4: Variational Autoencoders

4.1 Building and Training a Basic VAE

We implemented a Variational Autoencoder (VAE) to model the generative distribution of the MNIST dataset. The encoder was designed as a Multi-Layer Perceptron (MLP) with a single 400-neuron hidden layer, mapping the 28×28 input images to a 2-dimensional latent space (μ and $\log \sigma^2$). Using the reparameterization trick, we sampled from the approximate posterior to allow backpropagation through the stochastic node. The decoder, mirroring the encoder, mapped the 2D latent vectors back to the 784-dimensional pixel space. The model was trained for 15 epochs using the Adam optimizer, with the loss function defined as the negative Evidence Lower Bound (ELBO) – a combination of Binary Cross-Entropy (reconstruction loss) and KL Divergence. The training loss converged smoothly, decreasing from 189.18 to 152.90.

4.2 Analyzing VAE Performance

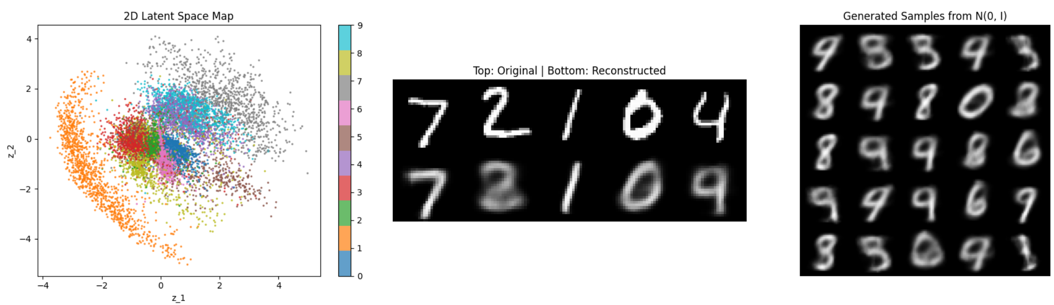

Figure 6: VAE Evaluation: 2D Latent Space Map (left), Reconstruction Quality (middle), and Generated Samples (right).

Latent Space Visualization: By setting the latent dimensionality to $d = 2$, we plotted the test set embeddings directly onto a 2D plane (Figure 6, left). The VAE successfully organized the latent space into a continuous manifold. Structurally distinct digits, such as '1' (orange), form highly isolated clusters. Digits with shared structural features, such as '4' and '9', overlap more heavily, demonstrating that the VAE has learned semantic visual relationships rather than just memorizing labels. **Reconstruction Quality:** The reconstructions (Figure 6, middle) preserve the global structure and distinct strokes of the original digits. However, they exhibit the characteristic blurriness associated with VAEs. This occurs because the model compresses the image into an extreme 2D bottleneck and is optimized using a pixel-wise Binary Cross-Entropy loss, which tends to average out fine details and sharp edges. **Generation of New Samples:** Sampling random vectors from the standard normal prior $\mathcal{N}(0, I)$ produced entirely new digit images (Figure 6, right). The samples reflect the diversity of the MNIST dataset, though some digits appear as ambiguous blends (e.g., a shape sitting between an '8' and a '3'). This is a direct consequence of sampling from the continuous interpolative spaces between distinct digit clusters in the prior.

4.3 Comparing with Doersch's VAE Implementation

We compared our custom implementation against the reference design outlined in Carl Doersch's tutorial.

- **Architecture:** Both implementations utilize a Multi-Layer Perceptron (MLP) architecture. Our model employs a single 400-neuron hidden layer for both the encoder and decoder, which aligns with the lightweight dense networks typically used in foundational VAE tutorials.
- **Output Distribution and Loss:** Our implementation exactly matches Doersch's approach to the output distribution. Doersch assumes a Bernoulli distribution for the MNIST pixel intensities, treating them as probabilities, and employs a sigmoid cross-entropy loss. We utilized a final Sigmoid activation in our decoder and computed the reconstruction loss using Binary Cross-Entropy (BCE).

- **Latent Dimensionality:** Doersch notes that VAEs are generally insensitive to the dimensionality of the latent space z, provided it is not excessively small or large. We deliberately chose a highly constrained latent dimension of $d = 2$ to allow for direct 2D visualization without secondary reduction techniques (like PCA or t-SNE). While this provided a clear visual map, this extreme compression is responsible for the slight drop in reconstruction sharpness compared to models with higher-dimensional latent spaces.
- **Results Comparison:** Our generated results share the exact same characteristics and failure cases noted by Doersch. While most generated samples are highly realistic, both our model and Doersch's implementation struggle with "in-between" digits. Because the VAE enforces a smooth, continuous latent space, sampling from the boundaries between two digit clusters naturally produces blurry, hybrid shapes that resemble a mix of multiple classes.